Draft version November 12, 2025
Typeset using LaTeX **twocolumn** style in AASTeX631

PageRank Analysis Using Iterative and Eigenvalue Methods on Directed Web Graphs

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ABSTRACT

In this project, we explore numerical methods for computing the PageRank vector, which ranks the relative importance of nodes in a directed network. The PageRank matrix is a stochastic transition matrix that encodes the probability of moving from one page to another. We first implement an iterative method that repeatedly applies the hyperlink matrix $\boldsymbol{H}$ to a uniform probability vector and observe its convergence to a unique steady-state solution. We then apply a direct method by solving for the dominant eigenvector, whose eigenvalue is $\lambda = 1$. Both approaches assume the matrix is ergodic (irreducible and aperiodic) to ensure convergence. When this condition fails, we introduce the "random surfer" model, which adds a damping factor to enable jumps to any node with fixed probability. This adjustment ensures convergence and models more realistic user behavior.

Keywords: stochastic matrix — hyperlink — directed graph — PageRank — eigenvalue — eigenvector — internet — search engine

1. INTRODUCTION

The PageRank algorithm, first introduced by Larry Page and Sergey Brin, was one of the first algorithms to rank webpages based on their link structure rather than solely content Brin & Page (1998). The algorithm models the web as a directed graph and simulates an "agent" who follows hyperlinks with some probability. Essentially, it tracks how often this "agent" would land on a particular page over an extended period, so pages that are visited more frequently are considered more important and thus receive a higher PageRank score Page et al. (1999).

This setup defines a Markov process where each page's importance is proportional to the importance of pages linking to it. The final PageRank vector corresponds to the steady-state distribution of this process and can be found either by iteration or by solving for the dominant eigenvector of the hyperlink matrix $\boldsymbol{H}$ Langville & Meyer (2009).

In this project, we first implemented PageRank using the difference equation $\boldsymbol{x}_{k+1} = \boldsymbol{H}x_k$, which resulted in the scores converging once the solution reached the maximum tolerance allowed. We then confirmed our result by solving for the eigenvector associated with the largest eigenvalue of the matrix $\boldsymbol{H}$. Our goal is to understand how the algorithm behaves under different network structures and how quickly it converges Langville & Meyer (2004).

2. DATA AND OBSERVATIONS

For an $n \times n$ hyperlink matrix $\boldsymbol{H}$ to be stochastic, each column must be a probability vector with non-negative entries that sum to 1. The matrix must also be ergodic, meaning it is both irreducible (every page can eventually be reached from any other page) and aperiodic (the system does not get trapped in cycles). This ensures that probability is conserved at each step of the Markov process and allows the system to converge to a unique steady-state solution. This condition is only satisfied if every page has at least one outgoing link. Otherwise, the corresponding column in $\boldsymbol{H}$ would contain all zeroes and violate the stochastic property Langville & Meyer (2009).

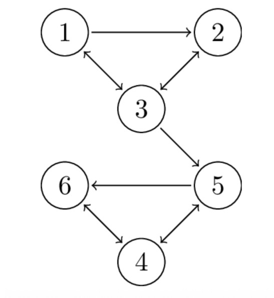

Figure 1. Example directed graph.

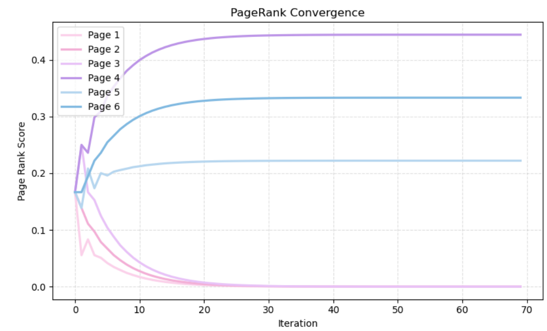

Figure 2. PageRank convergence for the graph in Figure 1.

To start, we define a 6-node directed graph where each page links to one or more pages, as shown in Figure 1. From this, we constructed the hyperlink matrix $\boldsymbol{H}$, where each non-zero entry $\boldsymbol{H}_{ij}$ is equal to $\frac{1}{|P_{ij}|}$, the reciprocal of the number of outbound links from page P_j. This matrix represents the probability that an agent on page j clicks through to page i:

$$\boldsymbol{H} = \begin{pmatrix} 0 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 1 \\ 0 & 0 & 1/3 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix} \tag{1}$$

We initialized the PageRank vector $\boldsymbol{x}_0$ with equal values and assumed that each page is equally likely to be visited:

$$\boldsymbol{x}_0 = \begin{pmatrix} 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \end{pmatrix} \tag{2}$$

We then applied the iterative update rule $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$ until the rank vector converged below a given tolerance to simulate an agent moving moving through the web graph via the links. We visualized the convergence of the PageRank vector by plotting each page's rank score for each iteration, which allowed us to track how the scores changed until they reached a steady state. Our results are shown in Figure 2.

The convergence of the iterative method is a direct consequence of the underlying theory of stochastic matrices. According to Theorem 4.9, if $\boldsymbol{H}$ is an $n \times n$ stochastic matrix and $\boldsymbol{x}_0$ is some initial vector, then the system defined by the difference equation $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$ converges to a unique steady-state vec Page et al. (1999). This limit corrresponds to the eigenvector $\boldsymbol{v}_1$ associated with eigenvalue $\lambda = 1$, since all the other eigenvalue components vanish as $k \to \infty$:

$$\boldsymbol{x}_{\text{equilibrium}} = \lim_{k \to \infty} \boldsymbol{H}^k \boldsymbol{x}_0 = c_1 \boldsymbol{v}_1 \tag{3}$$

Theorem 4.9 greatly simplifies the PageRank iterative process. Instead of repeatedly applying $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$, we can directly compute the dominant eigenvector of $\boldsymbol{H}$ corresponding to the eigenvalue $\lambda = 1$. This vector represents the steady-state solution and results in the same PageRank distribution as the limit of the iterative method Kamvar et al. (2003).

Both the iterative difference equation and the eigenvector method yield the same steady-state PageRank vector:

$$\boldsymbol{x}_{\text{converged}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 4/9 \\ 2/9 \\ 1/3 \end{pmatrix} \tag{4}$$

We applied the PageRank algorithm to a second web graph consisting of eight interconnected pages, as shown in Figure 3.

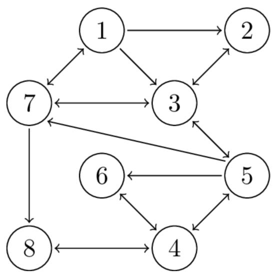

Figure 3. Example directed graph.

The corresponding hyperlink matrix $\boldsymbol{H}$ and initial uniform state vector $\boldsymbol{H}$ are defined below.

$$\boldsymbol{H} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1/3 & 0 \\ 1/3 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 1 & 0 & 0 & 1/4 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 1/4 & 1 & 0 & 1 \\ 0 & 0 & 1/3 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/4 & 0 & 0 & 0 \\ 1/3 & 0 & 1/3 & 0 & 1/4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 & 1/3 & 0 \end{pmatrix} \tag{5}$$

$$\boldsymbol{x}_0 = \begin{pmatrix} 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \end{pmatrix} \tag{6}$$

Using both the iterative update method and the dominant eigenvector approach, we computed the steady-state PageRank distribution. The convergence behavior of the iterative method is shown in Figure 4

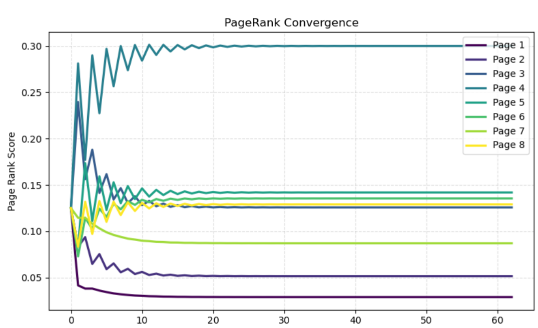

Figure 4. PageRank Convergence for the graph in Figure 3

Using both the iterative difference equation and the dominant eigenvector method, we arrive at the same steady-state PageRank vector:

$$\boldsymbol{x}_{\text{converged}} = \begin{pmatrix} 0.0290 \\ 0.0516 \\ 0.1258 \\ 0.3000 \\ 0.1419 \\ 0.1355 \\ 0.0871 \\ 0.1290 \end{pmatrix} \tag{7}$$

3. RESULTS

The distribution for the first web graph reflects the long-term behavior of an agent nagivating the web graph in Figure 1. Based on the values, we rank the pages in order of importance as follows:

$$\text{Page } 4 > \text{Page } 6 > \text{Page } 5 > \text{Page } 1 = \text{Page } 2 = \text{Page } 3$$

After a long enough time span, $\frac{4}{9}$ of users will end up in page 4, $\frac{2}{9}$ of users will end up in page 5, and $\frac{1}{3}$ of users will end up in page 6. Practically one will end up on pages 1, 2, and 3.

From the results of the final distribution for the graph in Figure 3, we rank the pages in order of importance:

$$\text{Page } 4 > \text{Page } 5 > \text{Page } 6 > \text{Page } 8 > \text{Page } 3$$
$$> \text{Page } 7 > \text{Page } 2 > \text{Page } 1$$

Page 4 is the most dominant, whereas Page 1 has the lowest influence in the network. Most people will end up on page 4 and the least amount of people will end up on page 1 in the limit of increasing time.

4. SUMMARY AND CONCLUSION

To summarize, we implemented two numerical approaches to compute the PageRank vector. The first was an iterative method that started from a uniform initial distribution $\boldsymbol{x}_0$ and repeatedly applied the update equation $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$ until the difference between successive iterations fell below a convergence threshold. The second approach used Theorem 4.9, which states that the steady-state distribution is the dominant eigenvector of the stochastic matrix $\boldsymbol{H}$. Both methods gave identical results, which confirms the accuracy of our implementation to in describing long-term internet behavior. We performed all computations in Python using `NumPy` and visualized convergence behavior with `matplotlib`.

One limitation of the PageRank model is the assumption that each transition is independent, meaning the current and previous pages visited do not influence the next page Langville & Meyer (2009). While this may have been reasonable when the internet was more decentralized, it does not reflect the behavior driven by modern recommendation algorithms which actively shape user navigation to maximize engagement and retention.

Another limitation is the the hyperlink matrix $\boldsymbol{H}$ must be ergodic for the iterative method to converge. In cases where this condition fails, such as with dangling nodes or disconnected components, we suggest introducing a pertubation to enforce ergodicity. This is achived by modifying the original matrix with a damping factor α to produce a new matrix:

$$\hat{\boldsymbol{H}} = (\boldsymbol{1} - \alpha)\boldsymbol{H} + \frac{\alpha}{n}\boldsymbol{J} \tag{8}$$

Here, $\boldsymbol{J}$ is a ones matrix and n is the total number of pages. This perturbation represents the the probability that a user may jump to any page and ensures the matrix remains stochastic. Thus, the random surfer model not only guaruantees convergence but also aligns more closely with realistic user behavior on the modern web Boldi et al. (2005).

ACKNOWLEDGEMENTS

We thank Professor Mitra for his guidance and instruction throughout the course, as well as the teaching assistants for their helpful feedback. We also acknowledge the Department of Physics for providing an excellent academic foundation to support this work. This work was completed as part of the requirements for C S 323E: Elements of Scientific Computing at the University of Texas at Austin.

REFERENCES

Boldi, P., Santini, M., & Vigna, S. 2005, in Proceedings of the 14th international conference on World Wide Web (New York, NY, USA: ACM), 557–566

Brin, S., & Page, L. 1998, Computer Networks, 30, 107. https://snap.stanford.edu/class/cs224w-readings/Brin98Anatomy.pdf

Kamvar, S. D., Haveliwala, T. H., Manning, C. D., & Golub, G. H. 2003, in Proceedings of the 12th International Conference on World Wide Web (New York, NY, USA: ACM), 261–270

Langville, A. N., & Meyer, C. D. 2004, Internet mathematics, 1, 335

—. 2009, Google's pagerank and beyond: The science of search engine rankings., 1st edn. (Princeton, N.J: Princeton University Press)

Page, L., Brin, S., Motwani, R., & Winograd, T. 1999, The PageRank Citation Ranking: Bringing Order to the Web., Technical Report 1999-66, Stanford InfoLab. http://ilpubs.stanford.edu:8090/422/